Basic Skill Level Exam

Name: _____ **Rank:** _____

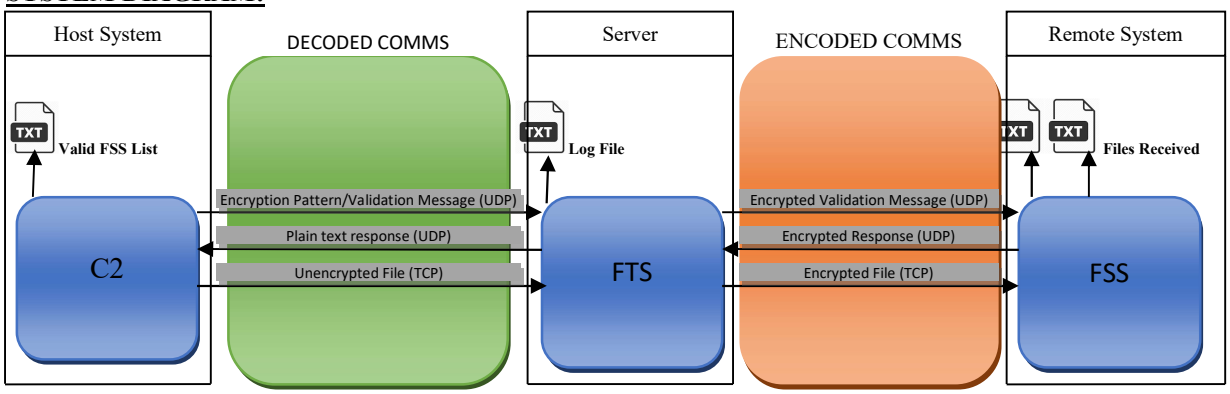
Unit: _____ **Date:** _____

Project Title: File Transfer Service

OVERVIEW:

Your organization is tasked with creating a File Transfer Service (FTS) that sends an encoded file to a designated File Storage Service (FSS). In order to test your developed FTS, your organization is required to create a simplified version of the command and control (C2) and File Storage Service (FSS) components that can handle the requirements outlined in this project.

SYSTEM DIAGRAM:



SYSTEM ARCHITECTURE:

The file transfer system consists of three programming components the C2, the FTS, and FSS, as well as a user defined communication protocol, and encode/decode functionality.

GRADING RUBRIC:

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Documentation				
User Manual (4%)	Is the user manual easy to understand?	1%		____ of 12%
	Does the user manual follow a logical sequence?	0.5%		
	Are the sections in the user manual Cleary defined and laid out?	0.5%		
	Does the document define all aspects of user interaction?	1%		
	Does the user manual provide a general overview of the project?	1%		
Test Report (4%)	Is the test report easy to understand?	1%		
	Does the test report document all test cases and their results?	1%		
	Does the test report provide detailed steps to repeat test?	1%		
	Does the test report provide screen shots where appropriate?	1%		
Project Challenges (%2)	Did the individual address any challenges or concerns with the project?	2%		
Grammar (%2)	Are all documents free of grammatical errors?	1%		
	Are all documents free of spelling errors?	1%		

Implementation and construction				
Source Code (25%)	Were comments descriptive, concise, and aid code readability?	4%		___ of 48%
	Was source code not written by the exam taker properly cited i.e. source location (to include web address or page number and book title), date taken, and author?	3%		
	Was memory managed appropriately: use of malloc/calloc/realloc to create memory, free to release memory, and free of memory leaks when ran with valgrind?	7%		
	Were variable names clear, concise and descriptive in regards to their use?	4%		
	Were all variables used?	4%		
	Were secure coding practices used?	3%		
Architecture 8%	Were effective and efficient data structures used?	2%		
	Did the program match the design specification?	2%		
	Was adequate input validation conducted?	2%		
	Was the code structured in a modular fashion?	2%		
Testing Methodology 5%	Was a test plan integrated into the design of the project?	1%		
	Was thorough testing conducted?	2%		
	Were all tests in the test report documentation repeatable?	1%		
	Did all tests produce the output as described in the documentation?	1%		
Version Control 5%	Was one branch for each feature constructed?	1%		
	Were tags created as releasable version of code were successfully integrated into the master branch?	1%		
	Was one branch per a feature used and merged to master as completed?	1%		
	No work was conducted in the master branch?	2%		
Modularity 5%	Were header and source files used to create modular code?	2%		
	Was code logically separated into multiple .h and .c files?	2%		
	Did the use of structures support the modularity of the code?	1%		
Execution				
Builds (10%)	When make is invoked does it compile without warnings when using –Wall command line option?	3%		___ of 40%
	When make is invoked does it build with the flags –Wall and –Werror?	6%		
	When make is invoked are all files placed in the specified directory?	1%		
Executes (10%)	Is the program fault tolerant when given bad input?	5%		
	Does input return the expected output?	2%		
	Does invalid input cause unexpected results and/or an error/crash?	3%		
Requirements (20%)	Are all requirements met?	6%		
	Was a multi-threaded server in C created as defined by the specification?	4%		
	Was a Command and Control element (C2) and File Storage Service (FSS) created as specified in the document?	5%		
	Is the transfer of a file from the C2 to the FTS and FTS to FSS supported as described in the document?	3%		
	Was encoding supported as specified in the document?	2%		
TOTAL		**Earned**		